

1 Conclusions

Chapter 1 framed this thesis as an end-to-end engineering pipeline (from a raw optical video archive to an uncertainty-aware design-triage tool) for marine multi-fuel pilot injection screening. The principal novelty of this thesis is not a new spray law or neural-network architecture, but an integrated, censoring-aware workflow that connects finite-FOV Mie observations, reduced-order trajectory targets, heteroscedastic surrogate learning, and geometric risk-oriented screening. This closing chapter synthesizes the core contributions, key scientific and empirical findings, evidence boundaries, and future validation priorities.

1.1 Novelty and Contributions

The primary contributions of this thesis are structured across three levels:

1. **Core Methodological Contribution:** An end-to-end censoring-aware learning chain that unifies finite-FOV observation auditing, Sigmoid-Gated Quarter-Root (SGQR) target construction, raw/teacher hybrid supervision, and heteroscedastic neural prediction. This formulation mitigates the risk of mistaking optical image window exit for physical spray saturation.
2. **Downstream Methodological Contribution:** A geometric screening interface that projects predicted penetration probability distributions into time-resolved wall-interception scores ($P_{\text{hit}}(t)$ and cumulative exposure E). This serves as a continuous geometric proxy for early-stage design triage ahead of computational fluid dynamics (CFD) simulation.
3. **Engineering Contribution:** An automated, CUDA-accelerated video processing and checkpointing pipeline capable of ingesting raw multi-hole Mie scattering archives (processing 2,992 recordings in approximately 72 minutes) into auditable plume-wise temporal observables.

1.2 Main Findings

The empirical evidence established in Chapters 3 through 5 supports six core findings, ordered by empirical strength:

1. **Prevalence of Field-of-View Censoring:** Across the cleaned optical dataset, 39.6% of plume trajectories exhibit spatial right-censoring at the camera field-of-view limit, demonstrating that right-censoring is a dominant structural characteristic of high-speed spray imaging archives rather than an isolated edge case.
2. **Target Construction Reliability:** The three-parameter SGQR target parameterization increases the trajectory fitting pass rate from 61.3% (under an unconstrained four-parameter baseline) to 90.0%, while maintaining median fit quality ($\text{RMSE} = 1.27 \text{ mm}$) on accepted fits. This confirms its

efficacy as a trajectory compressor and target generator, though it does not validate physical extrapolation beyond the camera window.

3. **Transient Pressure Scaling and Trajectory Collapse:** In short-duration pilot injections, the effective pressure-difference exponent (ΔP) is approximately 0.5 during early transient breakup before evolving toward classical asymptotic values. Applying a $\Delta P^{0.5}$ scaling exponent achieves an approximate $6.6\times$ tighter population collapse across operating conditions than default scaling, providing a physics-informed foundation for plume target aggregation.
4. **Surrogate Benchmarking against Empirical Baselines:** On the primary uncensored CDF evaluation protocol (540,367 points across 9 nozzle campaigns), the three-stage multilayer perceptron (MLP) surrogate achieves a mean RMSE of 4.19 mm and MAE of 2.98 mm. Within the bounds of this dataset, evaluation protocol, and calibrated baselines, this represents a 58.9% reduction in point-wise RMSE relative to refitted Hiroyasu–Arai (10.19 mm) and a 54.6% reduction relative to refitted Naber–Siebers (9.21 mm).
5. **Backbone Architecture Trade-offs (MLP vs. SVGP):** Comparative evaluation reveals that the Sparse Variational Gaussian Process (SVGP) benchmark is the superior point-wise and distributional predictor (RMSE = 4.09 mm; 1σ coverage 71.7% vs. nominal 68.3%). The MLP yields wider, more conservative in-domain prediction intervals ($1\sigma/2\sigma$ coverage 88.8%/99.0%) and facilitates modular integration with onset heads and knowledge-distillation chains; however, conservative coverage does not constitute a false-negative safety guarantee.
6. **Out-of-Distribution Transferability Limits:** Leave-one-nozzle-out (LONO) cross-validation indicates reasonable in-family transfer across evaluated nozzle geometries (RMSE $\approx$ 5.95 mm–7.00 mm excluding Nozzle 0). However, both surrogates exhibit significant error growth on the novel injector design fold (Nozzle 0; RMSE $\sim$ 27 mm–35 mm), highlighting the clear boundaries of inductive bias when transferring to unseen hardware families.

Table ?? summarizes these primary findings alongside their quantitative evidence and operational scope.

1.3 Evidence Boundary and Future Validation

The conclusions of this thesis are bounded by three key evidence gaps that require future experimental and computational validation:

- **Wide-FOV Extrapolation Validation:** The late-stage temporal extrapolation hypothesis of the SGQR target construction and knowledge distillation chain requires verification against wide-field single-plume optical recordings.

Finding / Theme	Quantitative Evidence	Operational Scope / Boundary
FOV Right-Censoring	39.6% of cleaned trajectories censored at FOV limit	Inherent to finite-window Mie archives
Target Fitting (SGQR)	Pass rate improved from 61.3% to 90.0% (median RMSE 1.27 mm)	Trajectory compressor; wide-FOV extrapolation unvalidated
Transient Pressure Scaling	$\Delta P^{0.5}$ yields $\sim 6.6\times$ tighter trajectory collapse	Early transient pilot injection regime
Empirical Baseline Gain	RMSE reduced by 58.9% (H-A) and 54.6% (N-S) on uncensored CDF	Scoped to measured dataset and refitted baselines
Model Trade-offs (MLP vs. SVGP)	SVGP lower RMSE (4.09 mm); MLP wider coverage (88.8% 1σ)	SVGP stronger regressor; MLP wider screening interval
OOD Hardware Transfer	LONO 4-fold RMSE 5.95 — 7.00 mm; Nozzle 0 RMSE ~ 27 — 35 mm	Reasonable in-family transfer; fails on novel hardware family

Table 1: Summary of primary findings, quantitative evidence, and scope boundaries established in this thesis.

- **Physical Impingement Ground Truth:** The probabilistic wall-interception metric $P_{\text{hit}}(t)$ acts solely as a geometric proximity proxy; validating whether elevated contact scores correspond to actual liquid droplet impingement or wall wetting requires matching high-temperature evaporative CFD or optical spray-wall interaction experiments.
- **Statistical Independence and Confidence Bands:** Because frame-wise observation points within a video trajectory are temporally autocorrelated, future statistical assessments should incorporate condition-level and nozzle-clustered hierarchical confidence bands.

1.4 Closing Statement

By establishing an end-to-end, censoring-aware workflow, this thesis demonstrates how archival optical Mie scattering recordings can be systematically repurposed into an uncertainty-aware pre-screening tool. The resulting methodology provides an auditable, data-efficient framework with the potential to narrow candidate nozzle geometries and operating conditions prior to resource-intensive CFD simulations and engine validation trials.